

BURSA, TurkeyWMO: **171160**Lat: **40.2308N**Lon: **29.0133E**Elev: **328**StdP: **14.52**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	28.8	21.1	16.0	31.6	23.4	17.9	33.6	20.7	56.7	17.9	57.3	2.8	60	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	94.4	71.9	91.7	71.6	89.5	70.9	75.4	89.3	73.9	87.4	72.7	85.8	8.4	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.0	115.6	83.4	69.5	109.6	82.0	68.1	104.6	80.4	39.0	89.1	37.7	87.6	36.5	85.9	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	14.3	12.2	19.7	101.0	4.6	3.5	16.4	103.5	13.8	105.5	11.2	107.5	7.9	110.0
			19.2	77.4	4.5	2.9	15.9	79.5	13.3	81.2	10.7	82.9	7.4	85.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.5	42.4	45.1	48.8	56.5	65.0	72.7	77.6	70.2	61.0	52.1	44.5	(6)	
(7)		DBStd	13.99	7.52	7.76	7.58	6.97	5.68	4.91	3.69	3.94	5.37	6.36	7.33	7.40	(7)
(8)		HDD50	827	256	172	117	19	0	0	0	0	3	57	203	(8)	
(9)		HDD65	3296	699	557	504	265	70	4	0	16	156	390	635	(9)	
(10)		CDD50	4303	21	36	81	215	466	681	845	854	607	343	121	33	(10)
(11)		CDD65	1298	0	1	2	11	71	235	380	389	173	31	4	1	(11)
(12)		CDH74	13109	0	4	27	174	754	2381	4048	3996	1460	243	22	0	(12)
(13)		CDH80	5367	0	0	3	37	206	944	1826	1803	495	51	2	0	(13)
(14)	Wind	WSAvg	4.6	4.8	5.2	5.1	4.5	4.5	5.2	5.3	4.1	3.4	3.6	4.7	(14)	
(15)	Precipitation	PrecAvg	27.6	3.5	2.8	2.8	2.6	1.8	1.5	0.6	0.7	1.6	2.8	3.0	4.4	(15)
(16)		PrecMax	52.3	8.3	7.0	5.8	5.9	5.0	5.3	2.8	3.3	5.2	15.6	5.6	9.6	(16)
(17)		PrecMin	17.6	0.3	0.3	0.2	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.3	(17)	
(18)		PrecStd	5.4	2.0	1.6	1.3	1.4	1.2	1.1	0.6	0.9	1.4	2.3	1.4	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.5	71.5	77.7	84.5	90.1	96.8	100.0	97.0	93.8	85.2	76.9	68.4	(19)
(20)		MCWB	53.5	55.4	59.1	64.3	68.6	72.6	73.8	73.3	70.3	65.6	59.8	56.4	(20)	
(21)		2%	DB	62.2	65.9	71.3	78.9	85.4	91.8	94.2	93.7	89.0	80.3	71.8	64.2	(21)
(22)		MCWB	50.6	53.0	55.8	61.8	66.9	70.6	72.4	72.2	68.9	64.9	58.6	54.4	(22)	
(23)		5%	DB	58.8	62.1	67.0	74.3	81.6	88.5	91.3	91.4	85.6	76.0	68.4	59.9	(23)
(24)		MCWB	49.9	51.6	54.3	60.1	65.2	69.6	72.0	72.1	68.5	63.5	57.0	51.9	(24)	
(25)		10%	DB	54.3	58.4	62.7	70.2	78.1	85.7	88.9	89.2	82.3	72.2	64.2	55.5	(25)
(26)		MCWB	47.7	50.0	52.2	58.1	63.8	68.6	71.1	71.3	67.0	61.8	55.9	50.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	57.6	61.5	66.3	71.0	75.5	77.2	77.6	73.8	69.7	62.9	57.6	(27)
(28)		MCDWB	63.4	67.8	75.0	79.7	85.6	90.7	92.1	92.2	86.9	80.1	72.2	65.1	(28)	
(29)		2%	WB	51.8	54.3	57.1	63.2	68.4	73.0	75.3	75.5	71.6	66.6	60.3	55.4	(29)
(30)		MCDWB	60.1	63.2	68.1	75.8	81.7	86.7	89.6	89.1	84.2	76.1	68.3	62.9	(30)	
(31)		5%	WB	49.9	52.1	54.6	60.8	66.5	71.4	73.8	74.1	70.1	64.7	58.2	52.7	(31)
(32)		MCDWB	57.5	60.7	64.9	72.0	78.6	84.8	87.7	87.2	81.4	73.3	65.7	58.7	(32)	
(33)		10%	WB	48.2	50.2	52.6	58.6	64.8	69.8	72.6	72.8	68.4	62.9	56.4	50.4	(33)
(34)		MCDWB	54.5	57.7	61.7	69.3	75.8	82.6	85.9	85.4	79.1	70.4	63.5	55.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.2	14.9	17.4	20.0	21.1	22.0	22.4	21.9	21.2	18.6	16.9	12.8	(35)	
(36)		5% DB	MCDBR	19.0	22.3	25.4	27.8	27.8	27.4	26.9	25.3	26.7	24.7	22.9	17.0	(36)
(37)		MCWBR	10.9	12.8	13.7	14.4	13.0	11.2	10.4	9.8	10.9	11.6	12.2	9.9	(37)	
(38)		5% WB	MCDBR	17.5	20.7	23.0	25.6	24.2	24.2	23.1	22.3	23.4	20.8	19.8	14.1	(38)
(39)		MCWBR	10.7	12.3	13.3	14.3	12.5	11.3	10.0	9.6	10.6	10.8	11.5	8.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.356	0.386	0.428	0.471	0.442	0.423	0.424	0.430	0.418	0.402	0.368	0.357	(40)	
(41)		taud	2.350	2.248	2.145	2.049	2.205	2.299	2.309	2.289	2.309	2.332	2.374	2.367	(41)	
(42)		Ebn at Noon	250	258	259	256	267	272	270	265	260	250	244	239	(42)	
(43)		Edh at Noon	28	36	44	52	45	41	40	40	37	33	27	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	522	752	1144	1561	1964	2242	2346	2047	1540	1026	688	484	(44)	
(45)		RadStd	73	110	111	176	158	183	81	113	117	87	86	72	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (14 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-176	N/A	+79	(47)